

Joel Emmanuel

Full-Stack Developer & Production Engineer

ejoe0035@gmail.com • github.com/JoelEmmanuelCloud • linkedin.com/in/joel-emmanuel-149708202 • joelemmanuel.dev

TECHNICAL SKILLS

Backend: Node.js, NestJS, Express.js, Python, Flask, Java, Spring Boot, REST APIs, GraphQL, Microservices, WebSockets
Production Engineering & SRE: Prometheus, Grafana, k6 Load Testing, Structured Logging, Health Checks, Runbooks, Failure Mode Analysis, Horizontal Scaling
Cloud & Infrastructure: AWS (EC2, S3, Lambda, RDS, SageMaker, Bedrock), Docker, Docker Compose, Nginx, GitHub Actions, CI/CD, DigitalOcean
Frontend & Databases: React.js, Next.js, TypeScript, Redux, Tailwind CSS | PostgreSQL, MongoDB, Redis, MySQL, Query Optimisation
AI & Blockchain: LangChain, AWS Bedrock, RAG Systems, LLMs, Vector Databases | Solidity, Web3.js, Lisk, Smart Contracts, Ethers.js

EXPERIENCE

Learnway Fullstack Developer · Remote	Dec 2024 – Present
- Building full-stack platform for Learnway blockchain education app on Lisk blockchain - Implementing scalable NestJS APIs for progress tracking and reward distribution - Integrating Lisk smart contracts for token-based educational rewards	
Invidia — AIQuickMatch Front-End Developer · Brazil (Remote)	Apr 2025 – Mar 2026
- Transitioned from Front-end QA to Developer within the same engagement - Built UI features with React 18, TypeScript, and TanStack Table - Integrated REST APIs with Keycloak SSO authentication and Zod validation	
Vivo Frontend Developer · Brazil (Remote)	2024 – 2025
- Developed React frontend for Vivo B2B fixed-sales e-commerce platform - Implemented Framework Brasil design system components and guidelines - Integrated with Node.js BFF and Java microservices decoupling layer	
Freedom Ghana Full-stack Developer · Ghana (Remote)	2023 – Dec 2024
- Built comprehensive ride-hailing platform with React and Node.js serving thousands of daily requests - Architected real-time ride matching with Socket.IO and Google Maps API - Deployed scalable AWS infrastructure with automated CI/CD pipelines	
Compass UOL Full Stack and AI/ML Developer · Brazil (Remote)	Feb 2023 – Present
- Developed e-commerce and enterprise applications with React, Spring Boot, and modern tech stack - Architected AI/ML solutions using AWS services including SageMaker and Bedrock - Built RAG system for Universidade de São Paulo with LangChain and vector databases	
Rewod Technologies Backend Developer · Nigeria	Oct 2022 – Mar 2023
- Developed internal tools using Python and Java Spring Boot - Created data processing pipelines handling millions of records daily	

KEY PROJECTS

Aegis Protocol – AI Agent Accountability Layer [live] ETHGlobal Open Agents, Ghent — May 2026 · 0G · Gensyn AXL · ENS · KeeperHub <i>TypeScript · NestJS · Solidity · 0G Storage/Compute · Gensyn AXL · ENS EIP-3668 · ERC-7857 · KeeperHub · React · Docker</i>	Lead Developer
- Built four Gensyn AXL nodes (Witness, Verifier, Propagator, Memory) over P2P mesh; attestations stored on 0G Storage with real merkle rootHashes - Deterministic 5-rule mandate engine with 0G Compute TEE notarization; verdicts recorded on AegisCourt.sol on 0G testnet - ENS *.aegis.eth subnames with live agent reputation readable from Ethereum via EIP-3668 CCIP-read — no bridge required	
Production URL Shortener – MLH PE Hackfest 2026 <i>Flask · PostgreSQL · Redis · Nginx · Docker · Prometheus · Grafana · k6 · GitHub Actions</i>	Production Engineer
- Redis caching reduced error rate from 1.68% to 1.28% (24% improvement) under 200 concurrent users - Health endpoint reports DB and cache status independently, enabling precise incident diagnosis - Failure mode documentation covering 6 failure scenarios with expected responses and recovery steps	

HACKATHONS

ETHGlobal Open Agents – Ghent	May 2026
Lead Developer · EthGlobal · OG · Gensyn · ENS · KeeperHub	
MLH Production Engineering Hackfest 2026	April 2026
Production Engineer · Major League Hacking (MLH)	
EthGlobal Unite DeFi Hackathon	July 25 – August 8, 2025
Blockchain Developer · EthGlobal, 1inch & Ethereum Developer Community	
EthGlobal x Avail: Building Unchained Apps	October 10 – 31, 2025
Web3 Developer · EthGlobal & Avail	

CERTIFICATIONS

AWS Certified Cloud Practitioner	2023-10
Amazon Web Services	
AWS Certified AI Practitioner	2024-03
Amazon Web Services	
Huawei Certified Network Associate	2023-06
Huawei Technologies	